KIEU STUDIO · GITHUB RADAR · EPISODE 14 · CHEAT SHEET

ghidra

The NSA's reverse-engineering tool: read programs without source code.

`github.com/NationalSecurityAgency/ghidra`

+16,685	96,650	Apache-2.0
stars in 30 days · ≈ 556 stars/day	total stars · GitHub API	license · created Mar 1, 2019

Ghidra is a software reverse-engineering framework created and maintained by the NSA Research Directorate, open source (Apache-2.0): disassembly, a decompiler, graphs, Java or Python scripting, many instruction sets and file formats, interactive or automated.

How it works in six steps

1. **Load an executable** — Windows · macOS · Linux
2. **Automatic analysis** — many instruction sets
3. **Disassembly and graphs** — see the structure
4. **Decompile** — machine code → readable
5. **Write scripts** — Java or Python
6. **Work as a team** — large reverse engineering

Quick start

```
# Install a JDK
JDK 25 64-bit

# Download the release
the ghidra_*.zip in Assets (not Source Code)

# Open the GUI
./ghidraRun (Windows: ghidraRun.bat)

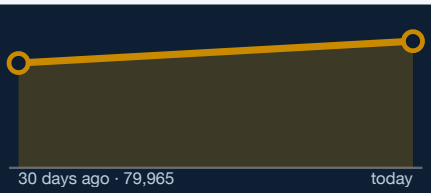
# Or use Python
./support/pyghidraRun

# Needs: JDK 25 64-bit; building from source also needs Gradle 9.1+, Python 3.9-3.15 and a C/C++ compiler
```

Key commands / features

- `./ghidraRun` — open Ghidra (ghidraRun.bat on Windows)
- `./support/pyghidraRun` — open PyGhidra
- `gradle buildGhidra` — build from source
- `gradle -I gradle/support/fetchDependencies.gradle` — fetch build dependencies

Growth



Two real points: the start of the window and today, axis from zero. The gain is as reported by GitHub Trending; we did not rebuild the star history.

Why this repo exists

- With only a compiled file you cannot see what a program does; reading machine instructions by eye is very slow.
- The NSA uses it to analyse malware and find vulnerabilities, and built it for teamwork on large reverse-engineering projects.

What to use it for

- Analyse executables without source code
- Decompile to follow the program flow
- Write Java or Python scripts to automate
- Work as a team on large reverse-engineering projects

Think twice before you use it

- The README warns that some versions have security vulnerabilities: read the Security Advisories before use.
- Needs JDK 25; use the official release file, not Source Code; do not unzip over an old install.
- Analysing unknown or malicious code: use an isolated virtual machine.
- The repo has over 2,100 open issues.

Sources: the repo's README, https://github.com/NationalSecurityAgency/ghidra. Stars: GitHub API; gains: GitHub Trending (as reported by Trending), snapshot Nov 1, 2026; stars measure attention, not quality: re-check before relying on them. Summarized in Kieu Studio's own words; commands copied from the README; no third-party logos or screenshots. **Disclosure:** drafted with AI assistance; the voice and presenter in the videos are AI-generated. Not investment advice, and not endorsed by Kieu Studio. **kieu.app**